

Chapter 1: Sets Theory, Relations and Functions - Formula Sheet

الفصل الأول: نظرية المجموعات، العلاقات والدوال

1. SETS THEORY | نظرية المجموعات

1.1 Basic Definitions | التعاريف الأساسية

English Term	Arabic Term	Symbol/Notation	Definition	Example
Set	مجموعة	$A = \{3,7,9,5\}$	Collection of objects	$A = \{1,2,3,4\}$
Element of	عنصر من	$x \in A$	x belongs to A	$5 \in A$
Not element of	ليس عنصراً من	$x \notin A$	x does not belong to A	$10 \notin A$
Empty Set	المجموعة الخالية	$\emptyset$	Set with no elements	$\emptyset = \{x \in \mathbb{R} \mid x^2 < 0\}$
Cardinality	العدد الأساسي	$\text{card}(A)$	Number of elements in A	$\text{card}(\{1,2,3\}) = 3$
Finite Set	مجموعة منتهية	-	Set with finite number of elements	$A = \{1,2,3\}$
Infinite Set	مجموعة لا منتهية	-	Set with infinite elements	$\mathbb{N} = \{1,2,3, \dots\}$
Set Builder	بناء المجموعة	$\{x \mid P(x)\}$	Set defined by property P	$\mathbb{N} = \{k \in \mathbb{Z} \mid k > 0\}$

Critical Note:

$$\emptyset \neq \{\emptyset\}$$

Empty set has 0 elements, $\{\emptyset\}$ has 1 element

1.2 Famous Number Sets | المجموعات العددية الشهيرة

Symbol	Name	Arabic Name	Description
$\mathbb{N}$	Natural numbers	الأعداد الطبيعية	$\{0,1,2,3, \dots\}$ or $\{1,2,3, \dots\}$
$\mathbb{Z}$	Integer numbers	الأعداد الصحيحة	$\{\dots, -2, -1, 0, 1, 2, \dots\}$
$\mathbb{Q}$	Rational numbers	الأعداد النسبية	$\{\frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0\}$
$\mathbb{R}$	Real numbers	الأعداد الحقيقية	All rational and irrational numbers
$\mathbb{C}$	Complex numbers	الأعداد المركبة	$\{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$

1.3 Set Equality | تساوي المجموعات

Concept	Definition	Formula
Equality	A and B have exactly the same elements	$A = B \Leftrightarrow \begin{cases} \forall x \in A, x \in B \\ \forall x \in B, x \in A \end{cases}$
Examples	-	$\{2,3,b\} = \{b,2,3\}$ but $\{2,3,a\} \neq \{2,4,a\}$

2. OPERATIONS ON SETS | العمليات على المجموعات

2.1 Inclusion | الاحتواء

Operation	Arabic Term	Symbol	Definition	Condition
Subset	مجموعة جزئية	$A \subseteq B$	A included in B	$\forall x, (x \in A) \Rightarrow (x \in B)$
Proper Subset	مجموعة جزئية حقيقية	$A \subset B$	A strictly included in B	$A \subseteq B$ and $A \neq B$

Example: $B = \{a, b, c, d\}, A = \{a, b\} \Rightarrow A \subset B$

2.2 Basic Set Operations | العمليات الأساسية

Operation	Arabic Term	Symbol	Definition
Intersection	التقاطع	$A \cap B$	Elements in both A and B $(x \in A \cap B) \Leftrightarrow (x \in A) \wedge (x \in B)$
Union	الاتحاد	$A \cup B$	Elements in A or B or both $(x \in A \cup B) \Leftrightarrow (x \in A) \vee (x \in B)$
Difference	الفرق	$A - B$ or $A \setminus B$	Elements in A but not in B $A - B = \{x \in A \wedge x \notin B\}$
Complement	المتمة	$\bar{A}$ or C_A^E	Elements in E but not in A $\bar{A} = C_A^E = \{x \in E \wedge x \notin A\}$

2.3 Advanced Operations | العمليات المتقدمة

Operation	Arabic Term	Symbol	Formula	Cardinality
Power Set	مجموعة الأجزاء	$\mathcal{P}(A)$	Set of all subsets of A	$\text{card}(\mathcal{P}(A)) = 2^{\text{card}(A)}$
Cartesian Product	الجداء الديكارتي	$A \times B$	Set of ordered pairs $A \times B = \{(x, y) \mid x \in A \wedge y \in B\}$	$\text{card}(A \times B) = \text{card}(A) \times \text{card}(B)$

3. PROPERTIES OF SET OPERATIONS | خصائص العمليات على المجموعات

3.1 Commutative Laws | قوانين التبديل

Operation	Formula
Intersection	$A \cap B = B \cap A$
Union	$A \cup B = B \cup A$

3.2 Associative Laws | قوانين التجميع

Operation	Formula
Intersection	$(A \cap B) \cap C = A \cap (B \cap C)$

Operation	Formula
Union	$(A \cup B) \cup C = A \cup (B \cup C)$

3.3 Distributive Laws | قوانين التوزيع

Law	Formula
Intersection distributes over Union	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Union distributes over Intersection	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

3.4 Identity Laws | قوانين المحايد

Operation	Formula
$A \cap \emptyset$	$A \cap \emptyset = \emptyset$
$A \cup \emptyset$	$A \cup \emptyset = A$
$A \cap A$	$A \cap A = A$
$A \cup A$	$A \cup A = A$

3.5 Complement Laws | قوانين المتممة

Law	Formula
Double Complement	$\overline{\overline{A}} = A$
De Morgan's Law 1	$C_{A \cap B}^E = C_A^E \cup C_B^E$ or $\overline{A \cap B} = \overline{A} \cup \overline{B}$
De Morgan's Law 2	$C_{A \cup B}^E = C_A^E \cap C_B^E$ or $\overline{A \cup B} = \overline{A} \cap \overline{B}$

3.6 Subset Equivalences | تكافؤات الاحتواء

Equivalence	Formula
Three Equivalent Conditions	$A \subseteq B \Leftrightarrow A \cap B = A \Leftrightarrow A \cup B = B$

3.7 Other Important Properties | خصائص أخرى مهمة

Property	Formula
Subset and Inclusion	$A \subseteq C \wedge B \subseteq C \Rightarrow (A \cap B \subseteq C) \wedge (A \cup B \subseteq C)$
Difference as Intersection	$A - B = A - (A \cap B)$
Difference as Union Minus	$A - B = (A \cup B) - B$

4. BINARY RELATIONS | العلاقات الثنائية

4.1 Definition | التعريف

Concept	Arabic Term	Definition
Binary Relation	علاقة ثنائية	$\mathcal{R}$ on set A is a subset of $A \times A$
Notation	-	$(a, b) \in \mathcal{R}$ or $a\mathcal{R}b$ means "a is related to b"

Concept	Arabic Term	Definition
General Form	-	$\mathcal{R} = \{(a, b) \in A \times A \mid \text{condition}\}$

Example: Let $A = \{7, 5, 6\}$, $B = \{5, 9, 8\}$ and $\mathcal{R}$ defined by $a > b$ Then $\mathcal{R} = \{(6, 5)\}$ and we write $6\mathcal{R}5$

4.2 Properties of Relations | خصائص العلاقات

Property	Arabic	Symbol	Definition	Example
Reflexive	انعكاسية	R	$\forall x \in A, x\mathcal{R}x$	$\leq$ on $\mathbb{N}$: $k \leq k$ ✓ $<$ on $\mathbb{N}$: NOT reflexive ✗
Symmetric	تماثلية	S	$\forall a, b \in A, a\mathcal{R}b \Rightarrow b\mathcal{R}a$	$k\mathcal{R}n \Leftrightarrow k + n$ is even
Anti-symmetric	لا تماثلية	AS	$\forall a, b \in A, (a\mathcal{R}b \wedge b\mathcal{R}a) \Rightarrow a = b$	$\leq$ on $\mathbb{R}$: $(a \leq b \wedge b \leq a) \Rightarrow a = b$
Transitive	تعدية	T	$\forall a, b, c \in A, (a\mathcal{R}b \wedge b\mathcal{R}c) \Rightarrow a\mathcal{R}c$	$<$ on $\mathbb{R}$: $(a < b \wedge b < c) \Rightarrow a < c$ Parallelism of lines

4.3 Special Types of Relations | أنواع خاصة من العلاقات

Type	Arabic Term	Required Properties	Formal Definition
Equivalence Relation	علاقة تكافؤ	Reflexive + Symmetric + Transitive	$\begin{cases} \forall x \in A: x\mathcal{R}x \\ \forall x, y \in A: x\mathcal{R}y \Rightarrow y\mathcal{R}x \\ \forall x, y, z \in A: (x\mathcal{R}y \wedge y\mathcal{R}z) \Rightarrow x\mathcal{R}z \end{cases}$
Ordering Relation	علاقة ترتيب	Reflexive + Anti-symmetric + Transitive	$\begin{cases} \forall x \in A: x\mathcal{R}x \\ \forall x, y \in A: (x\mathcal{R}y \wedge y\mathcal{R}x) \Rightarrow x = y \\ \forall x, y, z \in A: (x\mathcal{R}y \wedge y\mathcal{R}z) \Rightarrow x\mathcal{R}z \end{cases}$
Total Order	ترتيب كلي	Ordering + Comparability	$\forall x, y \in A: (x\mathcal{R}y) \vee (y\mathcal{R}x)$
Partial Order	ترتيب جزئي	Ordering but NOT Total	Not all pairs are comparable

4.4 Equivalence Classes | أصناف التكافؤ

Concept	Arabic Term	Notation	Definition
Equivalence Class of x	صنف التكافؤ	$[x]$	$[x] = \{y \in A \mid y\mathcal{R}x\}$
Properties	-	-	$[x] = [y] \Leftrightarrow x\mathcal{R}y$ $[x] \neq [y] \Rightarrow [x] \cap [y] = \emptyset$ - Equivalence classes partition A

5. FUNCTIONS | الدوال

5.1 Basic Definitions | التعاريف الأساسية

Term	Arabic Term	Notation	Definition
Function	دالة	$f: E \rightarrow F$	Rule associating each $x \in E$ with unique $y = f(x) \in F$
Mapping Notation	-	$x \mapsto f(x)$	Element x maps to f(x)
Domain	مجال التعريف	E	Set of input values (domain of departure)
Codomain	مجموعة الوصول	F	Set of possible output values

Term	Arabic Term	Notation	Definition
Image of E	صورة E	$f(E)$	$f(E) = \{f(x) \mid x \in E\}$
Pre-image	السلف	-	E is the pre-image of $f(E)$ under f

Example: $E = \{1,2,3\}$, $f(x) = 2x + 1 \Rightarrow F = \{3,5,7\}$

5.2 Types of Functions | أنواع الدوال

Type	Arabic Term	Definition	Alternative Statement
Injective (one-to-one)	تباينية	$\forall x_1, x_2 \in E, f(x_1) = f(x_2) \Rightarrow x_1 = x_2$	Each $y \in F$ has at most one pre-image
Surjective (onto)	غامرة (شاملة)	$\forall y \in F, \exists x \in E$ such that $y = f(x)$	Each $y \in F$ has at least one pre-image
Bijjective	تقابلية	Injective AND Surjective	Each $y \in F$ has exactly one pre-image

5.4 Function Composition | تركيب الدوال

Concept	Arabic Term	Notation	Definition
Composition	التركيب	$g \circ f$	For $f: E \rightarrow F$ and $g: F \rightarrow H$: $(g \circ f): E \rightarrow H$ $(g \circ f)(x) = g(f(x))$
Non-commutativity	عدم التبديل	-	$g \circ f \neq f \circ g$ (in general)

5.5 Identity and Inverse Functions | دالة الهوية والدالة العكسية

Function	Arabic Term	Definition	Properties
Identity Function	دالة الهوية	$\iota_E: E \rightarrow E$ $\iota_E(x) = x$	$\iota_E(x) = \iota_E(y) \Rightarrow x = y$
Inverse Function	الدالة العكسية	$f^{-1}: F \rightarrow E$	$\begin{cases} f^{-1} \circ f = \iota_E \\ f \circ f^{-1} = \iota_F \\ y = f(x) \Leftrightarrow x = f^{-1}(y) \end{cases}$
Existence Condition	شرط الوجود	-	f^{-1} exists $\Leftrightarrow f$ is bijective